

Project Executable Files

Date	04/07/2026
Team ID	Not Assigned
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	3 Marks

Project Executable Files

This section requires submission of all executable and deployable files for the Rising Waters flood prediction project. All files are properly organized, named, and uploaded to the GitHub repository. The evaluator can run or deploy the solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt (dependency file)	Yes
4	Database schema / seed files (if applicable)	NA
5	Environment configuration file (.env.example or similar)	NA
6	Deployed application URL (if hosted)	Yes
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	NA
9	Test files and test results	Partial — manual test results in Performance Testing document
10	Demo video or walkthrough (if required)	NA

Step 2: File / Folder Structure

The Rising Waters project follows a clean, organized structure as shown below:

```

Rising-Waters-Flood-Prediction/      <- Project Root
|
|-- app.py                          <- Flask web application (main file)
|-- train_model.py                  <- Model training script (Random Forest)
|-- predict.py                     <- Prediction helper module
|-- requirements.txt                <- Python dependencies
|-- README.md                      <- Setup and run instructions
|
|-- data/
|   |-- flood_dataset.xlsx          <- Training dataset (Kaggle)

```

```

|
|-- models/
|   |-- flood_model.pkl           <- Trained Random Forest model (Joblib)
|
|-- templates/
|   |-- index.html                <- Input form & prediction result page
(Jinja2)
|
|-- static/
|   |-- style.css                 <- Stylesheet

```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	https://rising-waters-flood-prediction-xbdj.onrender.com
Login Credentials (Demo)	No login required — application is publicly accessible without authentication
Platform / Hosting Provider	Render.com (Free Tier) — Python/Flask supported, HTTPS enforced, auto-deploy from GitHub
Repository Link	https://github.com/Pallaanilkumar655/Rising-Waters-Flood-Prediction
Demo Video Link	NA — Live application accessible directly at the deployed URL above

Step 4: Run Instructions

Provide clear, step-by-step instructions for the evaluator to run the project locally:

```

Pre-requisites:
- Python 3.8 or above
- pip package manager
- Git installed on system
- Internet connection (for package installation)

Step 1: Clone the GitHub repository
git clone https://github.com/Pallaanilkumar655/Rising-Waters-Flood-
Prediction.git

Step 2: Navigate to the project folder
cd Rising-Waters-Flood-Prediction

Step 3: Install all required dependencies
pip install -r requirements.txt

Step 4: Train the model (generates models/flood_model.pkl)
python train_model.py

Step 5: Start the Flask web application
python app.py

Step 6: Open browser and navigate to:

```

`http://127.0.0.1:5000`

Step 7: Access the live deployed application at:
`https://rising-waters-flood-prediction-xbdj.onrender.com`

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Render.com free-tier sleep mode — app goes to sleep after a period of inactivity	App wakes up automatically within 30–60 seconds when accessed. No action needed — simply wait for the page to load.
2	models/flood_model.pkl must be generated locally by running train_model.py before starting app.py	Resolved — the trained model file is committed to the GitHub repository, so it is already available.
3	No login or user authentication system implemented	Not required per project specifications. The application is intentionally publicly accessible for ease of use.
4	No live weather API integration — inputs must be entered manually	Planned as a future enhancement (see Scalability & Future Plan document).